Obtaining a support gap from fixed-target excess

A conditional reduction of the initial acquisition

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Abstract

The fixed-target formulation removes the need to estimate an unknown extremal score before using it as a coefficient. We now close a further initialization gap: a coarse cover of the third-moment interval suffices to obtain an actual central gap in the support. The main new estimate bounds a signed fourth-moment charge using signwise Cauchy–Schwarz. We give the derivation, the scalar acceptance rule, and the precise remaining dependencies. The resulting acquisition is conditional on the retained complete Fourier bounds; it is not a proof of the sharp Berry–Esseen conjecture or of the final local exclusion.

1 The point of the reduction

Fix $c = C_{\mathrm{E}} = (3+\sqrt{10})/(6\sqrt{2\pi})$. For a centered, variance-one law P, put $\rho = \mathbb{E}|X|^3$ and maximize the excess

$$a = \sqrt{n}\{F_{P^{*n}}(H) - \Phi(H/\sqrt{n})\} - c\rho.$$

Here H is a raw sum threshold. At a least failing index, the earlier fixed-target argument supplies an attained witness with $a \geq 0$ and sharp predecessor bounds. The contact coefficient is the known number c, even though the witness's ratio score equals $c + a/\rho$. This distinction removes the score-cap feedback from the initial entry. It does not remove attainment, contact, or the analytic Fourier error.

The earlier entry establishes

$$n \geq 42, \qquad 1 \leq \rho \leq 7361/4000, \qquad |X| \leq 96/25.$$

The support and moment statements require the fixed-target contact argument and the stated external structural inequality. They are premises of this note. The earlier 24-cell calculation obtained a first density license but left five cells outside the sufficient $D < 4$ regime. The improvement below closes that obstacle on the whole interval and, more significantly, obtains actual support geometry before using any central-gap refinement.

Write $\sigma = \sqrt{10}$ and let J be the actual weighted gap sum. The physical defect and conserved resource are

$$D = (3+\sigma)\rho - \gamma - 3J \geq 0, \qquad U = (n+H^2)a + \frac{n\phi(0)D}{6} + n^{3/2}\mathcal{L},$$

where $\gamma = \mathbb{E}X^3$ and $\mathcal{L} \geq 0$ is the contact loss. The raw remainder R_c satisfies

$$R_c = U + n(1 - e^{-H^2/(2n)})\left(c\rho - \frac{\phi(0)D}{6}\right). \tag{1}$$

A raw bound and a resource bound have different uses. A raw ceiling licenses a lower Gaussian density; a resource ceiling bounds nD. The computations below keep them separate at every step. In the two-pass statement we write the physical resource as S and its numerical ceiling as U.

What is proved here. The moment inequalities are proved for actual bounded laws, with both signs and zero atoms allowed. Their insertion into the retained scalar Fourier enclosure is checked by exact rational arithmetic on a complete closed cover. The resulting acquisition remains conditional on the contact/support premises and the full low- and high-frequency inequalities. No historical arithmetic success is used as a proof of those inequalities. The source and execution record accompanies this note.

2 Keep both signs in the fourth-moment charge

Let X be centered, with $\mathbb{E}X^2 = 1$ and $|X| \le R$ almost surely. Write

$$\rho = \mathbb{E}|X|^3, \quad \beta = \mathbb{E}X|X|, \quad \gamma = \mathbb{E}X^3, \qquad K_1 = \mathbb{E}[X|X|^3] - 3\beta - \tfrac{3}{2}\rho\gamma.$$

There is no central-gap or small-noise hypothesis. Atoms, including an atom at zero, are allowed. The aim is to bound the signed fourth-moment charge K_1 without combining incompatible extremal signs.

Put $m = \mathbb{E}X_+ = \mathbb{E}X_- > 0$ and $a_\pm = \mathbb{E}X_\pm^2$, $b_\pm = \mathbb{E}X_\pm^3$, $c_\pm = \mathbb{E}X_\pm^4$. Both $a_\pm$ are positive, and signwise Cauchy–Schwarz gives

$$a_\pm^2 \le m b_\pm, \qquad b_\pm^2 \le a_\pm c_\pm, \qquad c_\pm \le R b_\pm, \qquad 2m \le 1. \tag{2}$$

The last bound follows from $\mathbb{E}|X| \le \sqrt{\mathbb{E}X^2} = 1$. These are inequalities for one law, not a sufficiency claim for a moment relaxation.

Lemma 2.1 (A signed one-variable enclosure)**.** *Set*

$$C(r) = Rr - \tfrac{3}{2}r^2 - 3, \qquad a(r) = \tfrac{3}{2}r - R/2.$$

For either $\epsilon \in \{-1, 1\}$, let $h = 1 - \epsilon\beta$ and $\delta = \rho - \epsilon\gamma$. Then

$$h > 0, \qquad \delta \ge h^2, \qquad \epsilon K_1 \le C(\rho) + 3h + a(\rho)\delta - \frac{\delta^2}{2h}. \tag{3}$$

Proof. Take $\epsilon = 1$. Then $a_- = h/2$ and $b_- = \delta/2$. The first and last inequalities of (2) yield $h^2/4 \le m\delta/2 \le \delta/4$. The other two inequalities yield

$$c_- \ge \frac{\delta^2}{2h}, \qquad c_+ \le \frac{R}{2}(\rho + \gamma).$$

Insert these bounds into $K_1 = c_+ - c_- - 3\beta - 3\rho\gamma/2$, and substitute $\beta = 1 - h$, $\gamma = \rho - \delta$. This gives (3). Apply the same argument to $-X$ for $\epsilon = -1$. Its K_1 is $-K_1(X)$, while ρ and R are unchanged. □

For clarity, the absolute sign-moment cap also has a short direct proof. If $p_\pm = \mathbb{P}(\pm X > 0)$, Cauchy gives $m^2/a_\pm \le p_\pm$. Since $p_+ + p_- \le 1$, it follows that $m^2 \le a_+ a_- = (1 - \beta^2)/4$. Adding $a_\pm^2 \le m b_\pm$ therefore gives

$$\rho \ge \frac{1 + \beta^2}{\sqrt{1 - \beta^2}}. \tag{4}$$

In particular, a rational $b \in [0, 1)$ is a valid bound for $|\beta|$ throughout $\rho \le u$ whenever $(1 + b^2)^2 \ge u^2(1 - b^2)$. The right side of (4) is strictly increasing as a function of $|\beta|$.

3 A closed-cell bound with three scalar gates

Assume $\rho \in [l, u]$ and $|\beta| \le b < 1$, and put $h_* = 1 + b$. Ignoring the constraint $\delta \ge h^2$ and completing the square in (3) gives the universally valid preliminary bound

$$|K_1| \le B_R(\rho, b) := R\rho - \tfrac{3}{2}\rho^2 + 3b + \frac{1+b}{2}(\tfrac{3}{2}\rho - R/2)^2. \tag{5}$$

Indeed its remaining coefficient $3 + a(\rho)^2/2$ is positive, so $h \le h_*$ can be inserted after square completion. The maximum of the quadratic $B_R(r, b)$ on $[l, u]$ is at an endpoint, or at its vertex if the quadratic is concave and the vertex lies inside. Thus (5) already has a finite exact acceptance rule.

The constraint $\delta \ge h^2$ gives a stronger bound.

Theorem 3.1. *Suppose, in addition, that*

$$h_* \ge a(u), \tag{6}$$

$$3 + 2a(l)h_* - \tfrac{3}{2}h_*^2 \ge 0, \tag{7}$$

$$R - 3u + \tfrac{3}{2}h_*^2 \ge 0. \tag{8}$$

Then

$$|K_1| \le S(R, u, b) := Ru - \tfrac{3}{2}u^2 - 3 + 3h_* + (\tfrac{3}{2}u - R/2)h_*^2 - \tfrac{1}{2}h_*^3. \tag{9}$$

All cell endpoints are included.

Proof. For fixed $a = a(\rho)$, maximizing the right side of (3) over $\delta \ge h^2$ leaves $C(\rho) + \psi_a(h)$, where

$$\psi_a(h) = \begin{cases} h(3 + a^2/2), & 0 < h \le a, \\ 3h + ah^2 - h^3/2, & h \ge a. \end{cases}$$

The first branch exists only if $a > 0$ and is increasing. The second branch has derivative $3 + 2ah - 3h^2/2$, a concave quadratic in h. On $[0, h_*]$ its endpoint values are 3 and at least the left side of (7), because $a \ge a(l)$. It is therefore nonnegative throughout. The two branches agree at $h = a$, so ψ_a is increasing up to h_*. By (6), the value at h_* is the second branch. The resulting expression $C(\rho) + 3h_* + a(\rho)h_*^2 - h_*^3/2$ has derivative $R - 3\rho + 3h_*^2/2$, which is nonnegative on $[l, u]$ by (8). Insert $\rho = u$ to obtain (9). □

For the last recorded initial cell, the inputs are

$$R = 96/25, \quad l = 173303/96000, \quad u = 7361/4000, \quad b = 32078117/50000000.$$

The three gate margins are positive; the derivative margin in (7) exceeds 1.5445. Exact rational substitution gives

$$S(R, u, b) < \frac{3964267}{10^6} < \frac{397}{100}.$$

This improves the retained bound $|K_1| \le 5.8704858543$ on this cell. The same gates pass on cells 10–23 of the recorded 24-cell cover (and also cell 0, where the retained bound is better). Where a gate fails, use (5) or retain the earlier bound; the stronger formula is not extrapolated through a failed gate.

4 A short fifth-moment square and its unsigned use

Let $M_4 = \mathbb{E}X^4$ and $M_5 = \mathbb{E}|X|^5$. The identity

$$0 \le \mathbb{E}\{|X|^3(|X| - \rho)^2\} = M_5 - 2\rho M_4 + \rho^3$$

proves

$$\frac{\rho}{2}M_4 - \frac{1}{4}M_5 \le \frac{\rho^3}{4}. \tag{10}$$

This is sharp for the standardized symmetric three-point law on $\{-\rho, 0, \rho\}$ with masses $1/(2\rho^2)$ at each nonzero point, for every $\rho \ge 1$. A zero atom is therefore important in testing it. This inequality is already used in the retained signed code. The present step uses it in the earlier unsigned coefficient, replacing the weaker $8\rho^3/27$ term; no novelty claim is attached to the inequality.

Let $M \ge M_4$ and let $\Gamma(R, l) \ge 0$ be the retained negative-gap coefficient. In the retained coarse constant envelope, write

$$A_1 = \tfrac{3}{2}b(2u+1) + 3\Gamma(R,l)u + \tfrac{3}{8}u^2(u+1) + \tfrac{8}{27}u^3 - l/2, \quad A_2 = \tfrac{3}{2}b(2u+1) + 2uM.$$

The retained second branch is $A_2 - l/2$. With exactly the same gap and moment premises, the new constant is

$$A_{\text{new}} = \min\{A_1 - 5u^3/108,\ A_2 - l/2\}. \tag{11}$$

In particular its last-cell value is less than 10.406486, compared with 10.6950063919 previously. This step retains every other charge; it is not a new proof of the gap estimate represented by Γ.

5 Checks and limits

The exact checker reconstructs the 24 closed ρ cells, their rational moment caps and the retained scalar comparison formulas; it imports no archived program. It checks (4), all three gates where (9) is used, and the endpoint/vertex comparisons for (5). Failed monotonicity gates are kept as failures. Separate output columns distinguish the new universal bound from its minimum with the earlier conditional cap. No Fourier integral is evaluated by this check.

Actual-law controls include symmetric and skew Bernoulli laws, the sharp three-point example for (10), and the centered, variance-one law with atoms and masses

$$(-96/25, -1/2, 37/25), \qquad (325/22211, 11708/16533, 3650/13167).$$

For the last law,

$$\rho = 29884193/16458750 \in [l, u], \quad \beta = 70573/329175 > 0, \qquad \gamma = -23/1250 < 0.$$

Thus $\beta\gamma \ge 0$ is false even inside the last acquired moment cell. Also

$$K_1 = -102409783933/41146875000 < 0,$$

so $K_1\gamma \le 0$ is false there. Neither tempting sign shortcut is used.

The conclusions above are moment inequalities under an actual support bound. A raw-remainder or conserved-resource application must still supply the fixed-target contact/transfer identity, the acquired support, and the full low- and high-frequency estimates. No initial-acquisition theorem, local exclusion, saved certificate job, or global bound follows from these coefficient improvements alone.

6 Using the joint moment circle in initial acquisition

Let P be centered with variance one and finite third absolute moment. Write

$$\rho = \mathbb{E}|X|^3, \quad \gamma = \mathbb{E}X^3, \quad \beta = \mathbb{E}[X|X|], \quad m = \mathbb{E}X_+ = \mathbb{E}X_- > 0, \quad \sigma = \sqrt{10}.$$

All moments in this note belong to this one marginal. The moment-circle inequality is already present in the manuscript's Lemma `ms:physical-defect`; the new step here is its explicit inversion and use in the initial scalar feedback. No new independent bound for the Berry–Esseen constant is claimed.

Lemma 6.1 (Same-law moment constraint)**.** *For $a_\pm = \mathbb{E}X_\pm^2 = (1 \pm \beta)/2$ and $b_\pm = \mathbb{E}X_\pm^3 = (\rho \pm \gamma)/2$,*

$$mb_\pm \geq a_\pm^2, \qquad m^2 \leq a_+a_-, \qquad \gamma^2 + 4m^2 \leq \rho^2. \tag{12}$$

An atom at zero is allowed.

Proof. Both signs have positive probability, since the law is centered and nondegenerate. Cauchy–Schwarz applied on each sign to $X_\pm^{1/2}, X_\pm^{3/2}$ gives $a_\pm^2 \leq mb_\pm$. If $p_\pm = P(\pm X > 0)$, a second application gives $m^2 \leq p_\pm a_\pm$. Thus

$$m^2(1/a_+ + 1/a_-) \leq p_+ + p_- \leq 1.$$

As $a_+ + a_- = 1$, this proves $m^2 \leq a_+a_-$, without assigning a zero atom a false sign. Multiplying the first two bounds gives

$$b_+b_- \geq (a_+a_-)^2/m^2 \geq m^2.$$

Since $4b_+b_- = \rho^2 - \gamma^2$, the circle follows. In particular $2m \leq 1$ and $2m \leq \sqrt{\rho^2 - \gamma^2}$. □

For completeness, use the actual gap sum J of the manuscript and its conserved defect

$$D = (3 + \sigma)\rho - \gamma - 3J.$$

Its gap bound $J \leq \rho + 2m$ remains valid before a central gap is established. Indeed, on each gap lying on one side of zero, $(b-a)^2 \leq 2\int_a^b |x|\,dx$. Summing with its nonnegative gap weight and using $2\int |x|\tau(x)\,dx = \rho$ gives the one-sided contribution. If a central gap $(-A, B)$ exists, its additional contribution is $2mAB$. Since $(X + A)(X - B) \geq 0$ on support, $0 \leq \mathbb{E}[(X + A)(X - B)] = 1 - AB$. Thus $AB \leq 1$ and the bound follows. If zero belongs to support there is no central gap. Consequently

$$D \geq \sigma\rho - \gamma - 6m \geq \sigma\rho - \gamma - 3\sqrt{\rho^2 - \gamma^2}. \tag{13}$$

The first relation is an inequality, not an alternative definition of D.

Lemma 6.2 (Inverting a defect ceiling)**.** *Suppose $1 \leq l \leq \rho \leq u$, $D \leq d$ with $d \geq 0$, and $\sigma_l \leq \sqrt{10}$. Put $a_l = \sigma_l - d/l$. Define*

$$Y(a) = \begin{cases} -1, & a \leq -1, \\ \dfrac{a - 3\sqrt{10 - a^2}}{10}, & -1 < a \leq \sqrt{10}. \end{cases}$$

Then $\gamma/\rho \geq Y(a_l)$. A rational lower bound $y_l \leq Y(a_l)$ with $y_l \geq -1$ yields the valid cell bound

$$\gamma \geq \begin{cases} ly_l, & y_l \geq 0, \\ uy_l, & y_l < 0. \end{cases} \tag{14}$$

Proof. Let $y = \gamma/\rho \in [-1, 1]$. Dividing (13) by ρ gives

$$f(y) := y + 3\sqrt{1 - y^2} \geq \sigma - D/\rho \geq \sigma_l - d/l = a_l.$$

The case $a_l \leq -1$ imposes no new lower bound. On $[-1, 1/\sqrt{10}]$, f increases continuously from -1 to $\sqrt{10}$; its derivative in the interior is $1 - 3y/\sqrt{1-y^2}$ and changes sign only at $1/\sqrt{10}$. Solving $f(y) = a$ on this increasing branch gives $10y^2 - 2ay + a^2 - 9 = 0$ and the stated smaller root. It has $a - y \geq 0$, including $-1 \leq a \leq 0$, because there $\sqrt{10 - a^2} \geq -3a$. Hence squaring introduces no spurious branch. Every y below this root has $f(y) < a$, proving the lower bound. Finally multiplying $\gamma/\rho \geq y_l$ by the positive ρ requires the endpoint l for $y_l \geq 0$ and u for $y_l < 0$. □

In exact computation the square root in Y is rounded upward, so the negative root term rounds Y downward. The branch $a_l \leq -1$ keeps the whole interval. The implementation also intersects with the two previously justified bounds $\gamma \geq -g$ and $\gamma \geq \sigma_l l - 3 - d$. Thus

$$g_l = \max\{-g, \sigma_l l - 3 - d, \text{bound in (14)}\}, \tag{15}$$

$$\rho - \gamma \leq \min\{2u, (1 - \sigma_l)l + 3 + d, u - g_l\}. \tag{16}$$

These inequalities retain the same actual m, ρ and γ; they do not independently optimize over moments from different laws.

7 A shorter conditional entry statement

Let P be a centered variance-one fixed-target maximizing witness, $n \geq N = 42$, with actual support in $[-96/25, 96/25]$ and $1 \leq \rho = \mathbb{E}|X|^3 \leq 7361/4000$. Put $c = C_{\mathrm{E}}$, raw threshold H, $e = \exp[-H^2/(2n)]$ and $\phi_0 = (2\pi)^{-1/2}$. Assume the fixed-target contact/transfer identity

$$(n + H^2)a + nc\rho(1 - e) + n\phi_0 eD/6 + n^{3/2}L = \mathcal{R}_c(H), \qquad a, L, D \geq 0, \tag{17}$$

with the actual conserved defect $D = (3 + \sqrt{10})\rho - \gamma - 3J$. The complete retained low- and high-frequency estimates are also premises. They supply on each moment cell a raw quadratic enclosure

$$\mathcal{R}_c(H) \leq K_F + cA + f(c)|H| - q(c)H^2. \tag{18}$$

Every moment and gap in this statement belongs to the same law.

Use the 24 closed cells $[r_i, r_{i+1}]$ with $r_i = 1 + 3361i/96000$, $0 \leq i < 24$, and rational enclosures

$$c_- = 0.4097321837023 < c < c_+ = 0.4097321837025,$$

$$\phi_l = 997/2500 < \phi_0, \qquad \sigma_l = 3.162277660 < \sqrt{10}.$$

The signed Cauchy theorem supplies a gap-free cap K for $|K_1|$. The fifth-moment square replaces $8\rho^3/27$ by $\rho^3/4$ in the first unsigned constant branch. Concretely, for a cell $[l, u]$ with $|\beta| \leq b$, $|\gamma| \leq g$, $M_4 \leq M$, and retained gap coefficient $\Gamma \geq 0$, use

$$A(\Delta) = \min\{\tfrac{3}{2}b(2u + 1) + \tfrac{3}{2}\Gamma\Delta + \tfrac{3}{8}u^2(u + 1) + u^3/4 - l/2,$$
$$\tfrac{3}{2}b(2u + 1) + 2uM - l/2\}, \qquad f(t) = \qquad tK + v_H + s_F.$$

Here $K_F, v_H, s_F, M, b, g, \Gamma$ are exactly the freshly reconstructed first-cover quantities; no incomplete Fourier window is substituted.

7.1 The only phase price needed

A raw ceiling $\mathcal{R}_c \le T < Nc_-l$ licenses a positive density floor $\phi_- := \phi_l(1 - T/(Nc_-l))$ by (17). It also gives a quadratic price without any higher-degree maximization. Indeed, with $M_c = c\rho - \phi_0 D/6$,

$$eM_c = (1 + H^2/n)a + c\rho + \sqrt{n}L - \mathcal{R}_c/n \ge c\rho - T/n > 0.$$

Since $\exp(w) - 1 \ge w$ for $w = H^2/(2n)$,

$$n(1 - e)M_c \ge \frac{H^2}{2}(c\rho - T/n) \ge \eta H^2, \qquad \eta = (c_-l - T/N)/2 > 0.$$

Rearranging (17) consequently bounds the conserved resource

$$S := (n + H^2)a + n\phi_0 D/6 + n^{3/2}L \le \mathcal{R}_c - \eta H^2. \tag{19}$$

Thus a bound $S \le U$ implies $nD \le 6U/\phi_l$ and $D \le 6U/(N\phi_l)$. A resource ceiling is never used in place of T to license the density or to compute η.

7.2 Two ordered passes

In pass 1, take $\Delta = 2u$, $A = A(2u)$ and $q(t) = tl - g/5$. All divisors are strictly positive. For either pass, first compute

$$T_{\text{new}} = \min\left\{T_{\text{valid}}, \overline{\max_{t=c_-,c_+}\{K_F + tA + f(t)^2/(4q(t))\}}\right\}, \tag{20}$$

where T_{valid} is the retained first raw bound, also intersected with the preceding raw bound when available. A bar means upward rounding to a multiple of 10^{-10}; an underline below means downward rounding to that same grid. Form η from this new raw bound, then compute

$$U_{\text{new}} = \min\left\{T_{\text{new}}, U_{\text{valid}}, \overline{\max_{t=c_-,c_+}\{K_F + tA + f(t)^2/[4(q(t) + \eta)]\}}\right\}.$$

Omit U_{valid} on the first pass. The bound $S \le T_{\text{new}}$ is licensed by the positive M_c just proved. Set $d_{\text{new}} = \overline{6U_{\text{new}}/(N\phi_l)}$ and retain the minimum with any preceding defect bound.

For pass 2, use only pass 1's T_1, d_1. The same-law circle gives, without a central gap,

$$D \ge \sqrt{10}\rho - \gamma - 3\sqrt{\rho^2 - \gamma^2}.$$

Put $a_0 = \sigma_l - d_1/l$. If $a_0 \le -1$, take $y_l = -1$; otherwise obtain the rational lower bound $y_l \le (a_0 - 3\sqrt{10 - a_0^2})/10$ by rounding the square root upward. The circle inversion proves

$$g_l = \max\{-g, \sigma_l l - 3 - d_1, \underline{\min(ly_l, uy_l)}\} \le \gamma.$$

Its proof uses the increasing lower branch of $y + 3\sqrt{1 - y^2}$; it does not require a previously acquired support gap. Now use

$$\begin{aligned}
\Delta &= \min\{2u, (1 - \sigma_l)l + 3 + d_1, u - g_l\},\\
A &= \min\{A(2u), A(\Delta)\},\\
q(t) &= \max\{tl - g/5,\ tl + \tfrac{1}{2}\min(\phi_- g_l, (2/5)g_l)\}.
\end{aligned}$$

The actual gamma sign is retained. Both alternatives in q have slope l, so $q(t)$ is affine in t. For fixed $|H|$, the raw and resource quadratics are affine in t; their maxima occur at the two endpoints before square completion. All bounds use previous inputs before obtaining new conclusions. No central-gap estimate is used in either pass.

Theorem 7.1 (Conditional two-pass entry)**.** *Under the stated support, contact/transfer and complete Fourier premises, the two passes on all 24 closed cells prove*

$$\mathcal{R}_c \le 11.5164967977, \qquad S \le 9.9790315907,$$
$$D \le 3.5746638454 < 4, \qquad nD \le 150.1358815051.$$

Moreover the closed support of the witness has an actual central gap $(-A_0, B_0)$ *with* $A_0, B_0 > 0$.

Proof. The exact rational checker evaluates (20) and its resource counterpart on every cell, retaining closed shared endpoints:

pass	$\max T$	$\max U$	$\max d$	cells passing the gap test
1	13.2818901598	10.9330610254	3.9164138937	23
2	11.5164967977	9.9790315907	3.5746638454	24.

The second pass gives, in every cell, the strict margin

$$(\sigma_l - 1)l - d_2 \ge 526024602543/1600000000000 > 0.$$

It remains to justify what that test means. If zero belongs to the closed support, every complementary gap lies on one side of zero. For such a gap (a, b), $(b-a)^2 \le 2\int_a^b |x|\,dx$. Summing with its nonnegative zero-bias weight gives $J \le \rho$. Since $\gamma \le \rho$, this implies $D \ge (\sqrt{10} - 1)\rho \ge (\sigma_l - 1)l$, a contradiction. Thus zero is outside the closed support. Centering and positive variance place support on both signs; its nearest endpoints define the required central gap. □

The checker freshly computes the first-cover primitives and then exactly 48 cell passes. It does not import the higher-degree phase routine. Ten cells also have $D < 2$; fourteen do not. The theorem acquires an actual gap and a weighted defect bound, not the later much smaller defect, conditional-noise bounds or complete probability panels required for a local comparison. No existing certificate is removed by this result.

For this fixed cover the independent check also verifies that none of the retained minima improves the freshly completed quadratic ceilings in either pass. The fresh quadratics therefore suffice on their own: an earlier raw ceiling is not an additional mathematical premise of the two-pass result. The retained Fourier and moment estimates remain its premises.

8 An actual gap before the bound $D < 2$

The opposite-sign high-frequency event requires an actual gap around zero, not the particular sufficient bound $D < 2$. The stronger criterion below uses the moment cell. It does not assert that any numerical cell has acquired its hypotheses.

Let X have mean zero, variance one and finite third absolute moment ρ. Write $\gamma = \mathbb{E}X^3$, $m = \mathbb{E}X_+ = \mathbb{E}X_- > 0$, $\sigma = \sqrt{10}$ and

$$\tau(t) = \mathbb{E}[X\mathbf{1}_{\{X \ge t\}}], \quad J = \sum_{(u,v)} \tau_{u,v}(v-u)^2, \quad D = (3+\sigma)\rho - \gamma - 3J,$$

where the sum runs over bounded complementary intervals of the closed support and $\tau_{u,v}$ is the constant value of τ on (u, v).

Lemma 8.1 (Moment-sensitive actual-gap event)**.** *Suppose*

$$1 \le l \le \rho \le u, \quad D \le d, \quad d \ge 0, \quad 1 < \sigma_- < \sqrt{10}, \quad d < (\sigma_- - 1)l.$$

Define the positive numbers

$$P_0 = (\sigma_- - 1)l - d, \qquad S = l - \frac{d}{\sigma_- - 1}.$$

Then the actual closed support has a central gap $(-A, B)$, with $A, B > 0$, and

$$(A+B)^2 \geq 2S + \frac{2}{3}P_0 > 2S > 0. \tag{21}$$

For independent copies of this same law, the actual event $E = \{XX' < 0\}$ satisfies

$$\mathbb{P}(E) \geq \alpha := \frac{1}{2u^2}, \qquad |X - X'|^2 \geq 2S + \frac{2}{3}P_0 \quad \text{on } E, \qquad \mathbb{E}[(X-X')^2 \mid E] \leq \frac{2}{\alpha} = 4u^2. \tag{22}$$

In particular no premise $d < 2$ is needed for this event.

Proof. Signwise Tonelli gives $2\int |t|\tau(t)\,dt = \rho$. For a gap on one side of zero, $(v-u)^2 \leq 2\int_u^v |t|\,dt$. If zero belonged to the closed support, all its gaps would be of that kind, so $J \leq \rho$. As $\gamma \leq \rho$, this would imply $D \geq (\sigma-1)\rho \geq (\sigma_- - 1)l > d$, a contradiction. Closed support and both sign classes therefore give finite positive inner endpoints A, B. Endpoint atoms are allowed.

Put

$$T_+ = \mathbb{E}X_+^3 - mB^2, \quad T_- = \mathbb{E}X_-^3 - mA^2, \quad T = T_+ + T_-.$$

These are nonnegative. The central gap contributes $m(A+B)^2$ to J; write the remainder as J_o. For a positive outer gap, $(v-u)^2 \leq 2\int_u^v (t-B)\,dt$. Thus

$$J_o^+ \leq \mathbb{E}[X_+(X_+ - B)^2] = T_+ - 2B(\mathbb{E}X_+^2 - mB) \leq T_+.$$

The reflected argument gives $J_o^- \leq T_-$, hence $J_o \leq T$. Expanding D exactly gives

$$D = m\{\sqrt{\sigma+1}A - \sqrt{\sigma-1}B\}^2 + (\sigma-1)T_+ + (\sigma+1)T_- + 3(T - J_o).$$

Every term is nonnegative, so $T \leq D/(\sigma-1) \leq d/(\sigma_- - 1)$ and

$$m(A^2+B^2) = \rho - T \geq S.$$

For an additional product bound, summing the simpler gap inequality gives $J \leq \rho + 2mAB$: the central gap exceeds $2\int_{-A}^B |t|\,dt$ by exactly $2AB$. Consequently

$$6mAB \geq \sigma\rho - \gamma - D \geq (\sigma-1)\rho - D \geq P_0.$$

Since $m \leq 1/2$, combining the last two inequalities yields

$$(A+B)^2 \geq \frac{S + P_0/3}{m} \geq 2S + 2P_0/3.$$

There is no atom at zero. With $p = \mathbb{P}(X > 0)$ and $q = 1-p$, the actual event has probability $2pq$. Conditional Cauchy gives $1 = \mathbb{E}X^2 \geq m^2/p + m^2/q$, hence $pq \geq m^2$. Also $1 \leq \mathbb{E}|X|\,\mathbb{E}|X|^3 = 2m\rho$, while $2m = \mathbb{E}|X| \leq 1$. Therefore $2pq \geq 2m^2 \geq 1/(2\rho^2) \geq \alpha$. Opposite signs force $|X - X'| \geq A + B$. Finally $\mathbb{E}(X-X')^2 = 2$, so conditioning on E gives at most $2/\mathbb{P}(E) \leq 2/\alpha$. □

To use the complete high-frequency lemma, choose a positive rational d_0 with $d_0^2 \leq 2S + 2P_0/3$, use event probability lower bound $\alpha = 1/(2u^2)$ and conditional second-moment ceiling $4u^2$, and retain a valid fourth-moment ceiling for the same law. All the existing window, damping and index gates remain. This licenses no automatic use of the additional endpoint and moment conclusions bundled into the older $d < 2$ generic-event lemma.

In a finite ordered update, the new gap parameters must be obtained from the preceding proved defect ceiling. Every complete high-frequency bound then bounds the same actual Fourier integral. Its minimum with any previous complete bound is valid. If the same low-frequency triple is retained, the corresponding minima of the constant Fourier budgets are valid as well. Raw and conserved-resource bounds must still be kept distinct. This implication concerns the actual law throughout; it does not replace the law by an artificial two-point kernel.

9 Stronger bounds and the remaining numerical gap

The two-pass route above is sufficient for the initial support geometry. There are two bounded refinements. They give stronger quantitative information but are not needed to justify that route.

First, keep the full phase correction in (1). The appendix shows that its maximization over all real $n \geq 42$ reduces to a degree-six polynomial at the floor index. A short concavity test leaves one critical point, enclosed by rational bisection. Combined with the same moment coefficients, a single pass gives

$$T \leq 13.2818901598, \qquad U \leq 10.7965952605, \qquad D \leq 3.8675294672.$$

Every cell already has $D < (\sigma - 1)\rho$; the smallest rationally certified cell margin is greater than 0.03589975. Six further circle/density updates give $D \leq 3.3579256801$. The selected quadratic must itself satisfy its strict raw gate before the exponential estimate is used. A better bound obtained elsewhere by taking a minimum does not certify that condition for this quadratic.

Second, the actual central gap supplies a better symmetrization event. For independent X, X', use their opposite-sign event with probability at least $1/(2u^2)$, conditional second moment at most $4u^2$, and the proved distance bound in the preceding section. Re-evaluate the *complete* retained high-frequency bound, including all six pieces. Keep the smaller of this valid bound and the original universal bound. The lower-frequency estimate remains unchanged. Six bounded updates on the same 24 cells give

$$T \leq 9.3684595861, \qquad U \leq 8.3681296416, \qquad D \leq 2.9976105609, \qquad nD \leq 125.8996435547.$$

Sixteen cells satisfy $D < 2$; eight do not. All 24 eventually improve their high-frequency bound. None is omitted. The gap is established *before* its event is used, so this step does not assume its own conclusion.

Selected calculation	Upper bound for U	Upper bound for D	Gap cells
Two elementary passes	9.979032	3.574664	24/24
Full phase, one pass	10.796596	3.867530	24/24
Full phase, six circle updates	9.373986	3.357926	24/24
Full phase, six actual-gap updates	8.368130	2.997611	24/24

The decimal entries are outward upper bounds, not measured values of an extremizing law. The different rows have different dependencies and are not successive required stages of one proof.

Acceptance and its limits. The initial cells are $[1 + 3361i/96000, 1 + 3361(i+1)/96000]$, $0 \leq i < 24$. All include their endpoints. The checkers freshly reconstruct the first-cover constants from pinned source copies and use exact fractions, directed square roots and explicit continuous-domain maxima. They record both endpoints of the enclosure of c, complete high-frequency terms and the ordered previous-defect/new-raw/new-resource transitions. Separate checks reject missing, duplicated or reordered cells, invalid sign/moment gates, reversed root rounding, corrupted tables and an unlicensed phase envelope. Actual-law controls include Bernoulli, three- and four-point laws, zero atoms and tiny distant atoms; the last are rejected when they lie outside the acquired support domain.

Independent reviews rederive the moment and phase arguments. The combined polynomial review reconstructs the coefficients and certifies concavity by an independent Bernstein calculation. These checks do not independently prove the retained Fourier primitives or validate their entire dependency graph. A fresh arithmetic success proves only the specified scalar implication under those premises.

The retained initial acceptor asks for much more: its final targets include $D < 0.784$, raw remainder below 2.24, and further probability panels. Neither the two-pass result nor the stronger

refinements above establish that contract. The final signed support/noise domains, confinement of both deleted sums, strict count exclusion, zero-noise binomial comparison and independent finite base remain necessary. No existing certificate, driver or target has been removed or weakened.

10 How much of the proof does this replace?

The conceptual saving is substantial and quite specific. The old tail route first controls the maximizing ratio, then feeds that score interval through moment bounds, charges the excess, and improves the score cap. In the fixed-target route, the coefficient is c from the start. The nonnegative excess is retained on the left of the conservation identity. One therefore seeks physical information about the same law directly. The present result supplies an initial central gap by a short sequence of scalar estimates instead of assuming it as an input to a refinement.

For the selected short presentation, use the signwise moment bound, the fifth-moment square, the existing quadratic phase price, and one ordered moment-circle update. The stronger full-phase argument in the appendix is useful when a single pass is wanted, but need not be a compulsory step in that presentation. This is a choice between proofs of the same initial geometric conclusion; neither proof claims the later much smaller defect and noise parameters.

The current main manuscript has 158 pages. It already uses the least-counterexample reduction, so that earlier saving must not be counted again. The sections devoted to complete Fourier estimates, resource acquisition, confinement and count comparison occupy roughly pages 33–89; finite support and the finite-index analysis occupy roughly pages 101–156. These are measured heading spans, not disjoint estimates of deletable pages. Most of their conclusions remain necessary. Removing score interpolation alone saves a few local passages and a layer of reasoning, not half of the whole proof.

A large further reduction needs a theorem that replaces a substantial block, rather than a smaller numerical constant in each block. The most promising target is a uniform estimate for the *whole* fixed-target remainder, with its actual-gap baseline and Fourier error, that reaches the final moment/noise domain directly. A second is a strict count comparison valid over that whole domain and down to arbitrarily small positive noise. Those results could replace repeated acquisition or separate local witnesses. They are still open here. The independent finite base remains another essential task; moving its proof into a separate paper would shorten one document but not the overall proof.

In a consolidated manuscript, replace the old score-dependent derivation with the selected fixed-target derivation and prove each ingredient once. Do not append all successive research notes as additional required arguments. A readable organization is: finite base and first failure; attainment/contact/support; geometry and conservation; one complete Fourier engine; one ordered acquisition; count comparison; strict exclusion. The new gap theorem advances the acquisition step. It does not eliminate the Fourier engine, finite base or strict exclusion.

A Keeping the full phase correction

Let $c = C_{\mathrm{E}}$, $h = |H|$, $w = h^2/(2n)$ and $e = e^{-w}$. Write the conserved identity as

$$R = U + n(1-e)\left(c\rho - \frac{\phi(0)D}{6}\right), \qquad U = (n+H^2)a + \frac{n\phi(0)D}{6} + n^{3/2}\mathcal{L}.$$

Assume $a, D, \mathcal{L} \geq 0$, $n \geq N > 0$, and a valid positive lower bound $b \leq c\rho$. Since $U \geq n\phi(0)D/6$, this gives

$$U \leq nb + e^{h^2/(2n)}(R - nb). \tag{23}$$

Unlike the earlier constant phase price, this estimate keeps the dependence of R on the threshold. It requires no small-defect assumption.

Lemma A.1 (Only the floor and infinite-index limit are needed)**.** *For any real r, $b > 0$, $h \geq 0$ and $n \geq N > 0$,*

$$nb + e^{h^2/(2n)}(r - nb) \leq \max\left\{r - \frac{bh^2}{2},\ Nb + e^{h^2/(2N)}(r - Nb)\right\}. \tag{24}$$

This is an inequality for every real index $n \geq N$, hence for all integers.

Proof. The case $h = 0$ is immediate. Put $a_0 = h^2/2 > 0$, $x = 1/n$, and extend

$$f(x) = re^{a_0 x} - b\frac{e^{a_0 x} - 1}{x}$$

continuously to $x = 0$, with $f(0) = r - ba_0$. Its derivative has the sign of

$$r - ba_0\, j(a_0 x), \qquad j(z) = \frac{z - 1 + e^{-z}}{z^2} = \int_0^1 (1 - s)e^{-zs}\, ds.$$

Indeed $f'(x) = a_0 e^{a_0 x}[r - ba_0 j(a_0 x)]$. The integral shows that j strictly decreases on $[0, \infty)$. Thus the derivative can change sign only from negative to positive. There is no interior maximum on $[0, 1/N]$; the two endpoint values give (24). No monotonicity of the full expression in n is presumed. □

The infinite-index endpoint can be dispensed with when an upper bound on a nonnegative resource is wanted. If $f(0) = r - ba_0 > 0$, then $r > ba_0$ and $j \leq 1/2$ make $f' > 0$, so $f(1/N) > f(0)$. Otherwise $f(0) \leq 0$. Thus the right side of (24) is at most the positive part of its finite endpoint. Keeping that positive part is essential; the expression need not be monotone in n when its values are negative.

Suppose now a valid raw envelope is

$$R \leq r(h) = A + Fh - qh^2 \leq T < Nb, \qquad F \geq 0, \quad q > 0.$$

The first inequality in (23) is increasing in R, so replace it by $r(h)$. At the finite endpoint, $r(h) - Nb < 0$. Consequently the lower Taylor bound $e^z \geq 1 + z + z^2/2$ gives an *upper* resource bound. Define

$$P_\infty(h) = A + Fh - (q + b/2)h^2,$$
$$P_N(h) = r(h) + (r(h) - Nb)\left(\frac{h^2}{2N} + \frac{h^4}{8N^2}\right).$$

Then, uniformly for all $n \geq N$ and all thresholds,

$$U \leq \max\{0, P_N(h)\}. \tag{25}$$

The infinite-index polynomial has exact maximum $A + F^2/[4(q + b/2)]$ and remains a useful comparison. The finite-index polynomial has degree six and negative leading coefficient. On $r(h) < 0$ the nonnegative raw conserved quantity cannot occur, so a certified finite interval containing $\{h \geq 0 : r(h) \geq 0\}$ suffices.

There is a simple analytic test for a unique maximum; no positivity subdivision is necessary. Put

$$z = \frac{F^2}{8q^2 N}.$$

If $z + z^2 < 1$, then P_N is strictly concave on $[0, \infty)$. Indeed, writing $v(h) = h^2/(2N) + h^4/(8N^2)$ gives

$$P_N'' = -2q(1 + v) + 2(F - 2qh)v' + (r - Nb)v''.$$

Here $v'' > 0$ and $r - Nb < 0$. For $h \geq F/(2q)$ every term is nonpositive and the first is negative. Before that point, $h^2/(2N) \leq z$ and $h(F - 2qh) \leq F^2/(8q)$, whence

$$P_N'' \leq -2q + \frac{F^2}{4qN}(1 + z) = -2q(1 - z - z^2) < 0.$$

Since $P_N'(0) = F \geq 0$, its unique maximum is attained at $h = 0$ when $F = 0$, and otherwise at its unique positive stationary point. Rational bisection of the derivative followed by an outward monomial bound encloses that value. The check of $z + z^2 < 1$ is a scalar inequality, not an assumption of sampled concavity. The old 24-cell coefficients already have $z < 0.325$; decreasing F or increasing q preserves this gate.

For a formal coefficient interval containing c, take A, F, q, b affine in that coefficient, holding the same-law moment bounds fixed. Each of the two displayed polynomials is then affine before maximization over h. Their suprema, and their maximum, are convex in the coefficient; its two endpoints again suffice. The strict gate $T < Nb$ must hold for the chosen envelopes before reversing the exponential estimate.

This is a stronger way to bound U, not a new raw-remainder estimate. The raw ceiling T still supplies any earlier density license. Neither the low/high Fourier estimates nor the finite base are removed by this calculation.

B Provenance and review scope

The preceding fixed-target companion supplies attainment, contact and conservation. The entry note supplies the initial support argument and the first scalar cover. Exact source paths are listed in the accompanying README. The current directory contains the complete new TeX arguments, exact checkers, independent review reports and a portable source archive. The measured manuscript inventory and line-level dependency map are in `compression-map.md`. It distinguishes source removal from a replacement's use in a new proof composition.

AI systems were used to develop candidate arguments, write and revise this note, implement exact checks, and conduct separate adversarial reviews. Agreement between AI reviews is not a mathematical validation. The proof arguments and executable checks are supplied for the authors' and other readers' scrutiny. Remaining mathematical premises are stated explicitly; this document is a research deliverable, not a claim of a complete proof.